ANDROID PROGRAMMING PRACTICAL 04

Module 1 · Create a Fragment and demonstrate its lifecycle

Aim

Create and integrate a Fragment and demonstrate Fragment lifecycle methods.

Project name: FragmentLifecycleDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create an Empty Views Activity project and ensure AndroidX Fragment support is available.

2. Create DemoFragment extending androidx.fragment.app.Fragment and add fragment_demo.xml.

3. Inflate fragment_demo.xml in onCreateView(); add Log.d statements to onAttach(), onCreate(), onCreateView(), onViewCreated(), onStart(), onResume(), onPause(), onStop(), and onDestroyView() as appropriate.

4. Add a FragmentContainerView or container in activity_main.xml.

5. In MainActivity, use getSupportFragmentManager().beginTransaction() to add DemoFragment to the container.

6. Run the app and observe the FragmentLifecycle tag in Logcat.

7. Navigate away and back to observe view and lifecycle callbacks.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The fragment appears in its host activity and lifecycle messages can be observed.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.